

CS60216: Safe Gen-AI

TUTORIAL 1

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Contents

- **Chapter 1:** Hugging Face
- **Chapter 2:** PEFT, LoRA, Quantization, and QLoRA
- **Chapter 3:** DPO, and PPO
- **Chapter 4:** Model Arithmetic, Model Merging Toolkit, PyTorch Hooks

Chapter 1

What is Hugging Face?

The screenshot displays the Hugging Face website. At the top, the navigation bar includes the Hugging Face logo, a search bar, and links for Models, Datasets, Spaces, Community, Docs, Enterprise, Pricing, Log In, and Sign Up. The 'Models' link is highlighted with a red box, and the 'Docs' link is highlighted with a blue box. Below the navigation bar, a large banner features the Hugging Face logo and the text 'The AI community building the future.' followed by 'The platform where the machine learning community collaborates on models, datasets, and applications.' and buttons for 'Explore AI Apps' and 'Browse 2M+ models'. To the right, a list of models is shown, including 'meta-llama/llama-2-70b', 'stabilityai/stable-diffusion-xl-base-0.9', 'openchat/openchat', 'lillyasviel/ControlNet-v1-1', 'cerspense/zeroscope_v2_XL', 'meta-llama/llama-2-13b', 'tiiuae/falcon-40b-instruct', 'WizardLM/WizardCoder-15B-V1.0', 'CompVis/stable-diffusion-v1-4', 'stabilityai/stable-diffusion-2-1', and 'Salesforce/xgen-7b-8k-inst'.

Hugging Face Search models, datasets, users...

Models Datasets Spaces Community Docs Enterprise Pricing Log In Sign Up

The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

Explore AI Apps or Browse 2M+ models

Tasks Libraries Datasets Languages Licenses Other

Filter Tasks by name

Multimodal

- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Image-to-Image
- Unconditional Image Generation
- Video Classification
- Zero-Shot Image Classification

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Conversational
- Text Generation
- Text2Text Generation
- Sentence Similarity

Audio

- Text-to-Speech
- Automatic Speech Recognition
- Audio-to-Audio
- Audio Classification
- Voice Activity Detection

Tabular

- Tabular Classification
- Tabular Regression

Reinforcement Learning

- Reinforcement Learning
- Robotics

Models 469,541 Filter by name

- meta-llama/llama-2-70b
Text Generation • Updated 4 days ago • ± 25.2k • ♥ 64
- stabilityai/stable-diffusion-xl-base-0.9
Updated 6 days ago • ± 2.01k • ♥ 393
- openchat/openchat
Text Generation • Updated 2 days ago • ± 1.3k • ♥ 136
- lillyasviel/ControlNet-v1-1
Updated Apr 26 • ± 1.67k
- cerspense/zeroscope_v2_XL
Updated 3 days ago • ± 2.66k • ♥ 334
- meta-llama/llama-2-13b
Text Generation • Updated 4 days ago • ± 328 • ♥ 64
- tiiuae/falcon-40b-instruct
Text Generation • Updated 27 days ago • ± 288k • ♥ 899
- WizardLM/WizardCoder-15B-V1.0
Text Generation • Updated 3 days ago • ± 12.5k • ♥ 332
- CompVis/stable-diffusion-v1-4
Text-to-Image • Updated about 17 hours ago • ± 448k • ♥ 5.72k
- stabilityai/stable-diffusion-2-1
Text-to-Image • Updated about 17 hours ago • ± 782k • ♥ 2.81k
- Salesforce/xgen-7b-8k-inst
Text Generation • Updated 4 days ago • ± 6.18k • ♥ 57

Libraries in Hugging Face

Documentation

Ask HuggingChat

Hub & Client Libraries

Hub
Host Git-based models, datasets, and Spaces on the HF Hub

Hub Python Library
Python client to interact with the Hugging Face Hub

Huggingface.js
JavaScript libraries for Hugging Face with built-in TS types

Tasks
Explore demos, models, and datasets for any ML tasks

Dataset viewer
API for metadata, stats, and content of HF Hub datasets

Deployment & Inference

Inference Providers
Call 200k+ models hosted by our 10+ Inference partners

Inference Endpoints (dedicated)
Deploy models on dedicated & fully managed infrastructure on HF

Deploying on AWS
Train/deploy models from Hugging Face to AWS with DLCs

Text Generation Inference
Serve language models with TGI optimized toolkit

Text Embeddings Inference
Serve embeddings models with TEI optimized toolkit

Microsoft Azure
Deploy Hugging Face models on Microsoft Azure

Google Cloud
Train and Deploy Hugging Face models on Google Cloud

Core ML Libraries

Transformers
State-of-the-art AI models for PyTorch

Diffusers
State-of-the-art Diffusion models in PyTorch

Datasets
Access & share datasets for any ML tasks

Transformers.js
State-of-the-art ML running directly in your browser

Tokenizers
Fast tokenizers optimized for research & production

Evaluate
Evaluate and compare models performance

timm
State-of-the-art vision models: layers, optimizers, and utilities

Sentence Transformers
Embeddings, Retrieval, and Reranking

Kernels
Load and run compute kernels from the Hugging Face Hub

Training & Optimization

PEFT
Parameter-efficient fine-tuning for large language models

Accelerate
Train PyTorch models with multi-GPU, mixed precision

Optimum
Optimize HF Transformers for faster training/inference

AWS Trainium & Inferentia
Train/deploy Transformers/Diffusers on AWS

Google TPU
Train and deploy models on Google TPUs via Optimum.

TRL
Train transformers LLMs with reinforcement learning

Safetensors
Safe way to store/distribute neural network weights

Bitsandbytes
Optimize and quantize models with bitsandbytes

Lighteval
All-in-one toolkit to evaluate LLMs across multiple backends

Collaboration & Extras

Gradio
Build ML demos and web apps with a few lines of Python

Trackio
A lightweight, local-first, and free experiment tracking Python library

smolagents
Small library to build great agents in Python

LeRobot
Making AI for Robotics more accessible with end-to-end learning

Reachy Mini
Open-source expressive robot SDK for hackers and AI builders

AutoTrain
Automated API and UI for seamless model training

Chat UI
Open source chat frontend powering HuggingChat

Leaderboards
Create custom Leaderboards on Hugging Face

Argilla
Collaboration tool for building high-quality datasets

Distilabel
Framework for synthetic data generation and AI feedback

Community

Blog Learn Discord Forum GitHub

Transformers: Detailed

<https://huggingface.co/learn/llm-course/en/chapter2/2>

Models - <https://huggingface.co/docs/transformers/index>

How are Models Stored?

Let us analyze one model

<https://huggingface.co/EleutherAI/pythia-14m/tree/main>

How to Generate Access Tokens? You will need it :)

It's an exercise for you :)

Example: How to Train and Evaluate Models? (Full Fine-Tuning)

<https://huggingface.co/learn/llm-course/en/chapter3/4>

Using Trainer API (an exercise for you) -

<https://huggingface.co/learn/llm-course/en/chapter3/3>

Chapter 2

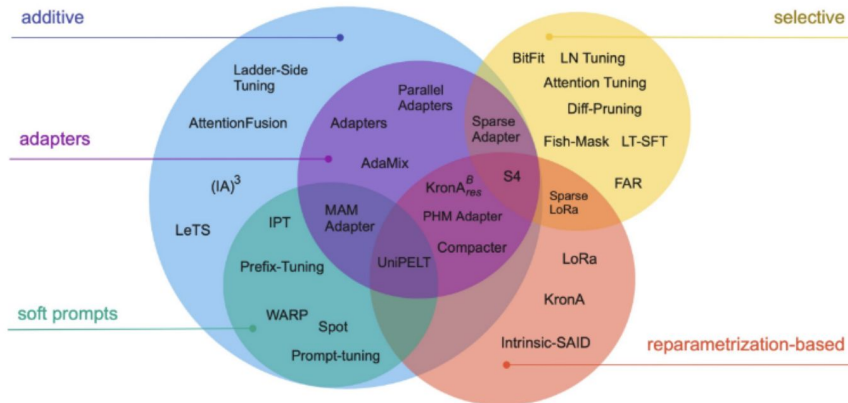
Why is Full Fine-Tuning a Problem?

There are many problems, but here are a few:

- Not everyone is (too) rich to full fine-tune models with even more than 10B parameters. ***You will know it soon :)***
- Student assignments must not be a function of extensive fine-tuning time. ***But TAs will not follow this rule, as usual :)***
- ***Humans are smart*** – we are less prone to ***Catastrophic Forgetting*** compared to GenAI models.

PEFT

- Update a small, task-specific set of parameters, keeping the base model weights mostly fixed.
- Examples include adapters, **LoRA**, **QLoRA**, prefix-tuning, etc



For you to explore -

<https://huggingface.co/blog/samuellimabraz/peft-methods>

Data Formats in PyTorch, LoRA, Quantization, QLoRA

Quantization (for you to explore) -

<https://medium.com/@abonia/llm-series-quantization-overview-1b37c560946b>

<https://abvijaykumar.medium.com/fine-tuning-llm-parameter-efficient-fine-tuning-peft-lora-qlora-part-1-571a472612c4>

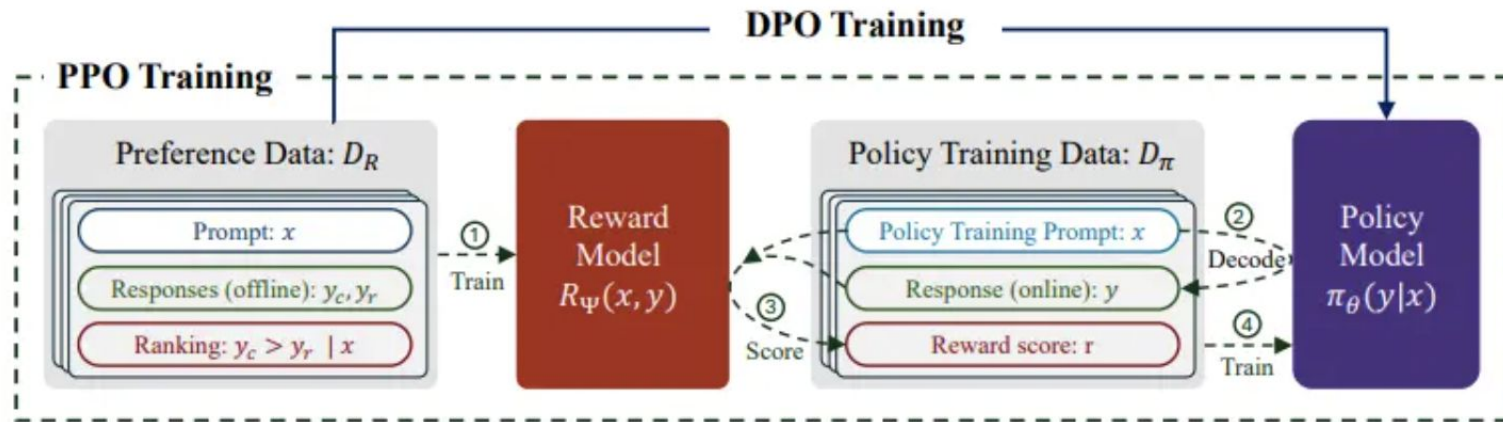
Sample Codes

LoRA - It's an exercise for you :)

QLoRA

Chapter 3

DPO vs PPO



<https://huggingface.co/learn/deep-rl-course/unit8/introduction>

For you to explore -

<https://huggingface.co/blog/NormalUhr/rlhf-pipeline>

Documentation: PPO

https://huggingface.co/docs/trl/v0.8.6/en/ppo_trainer

Documentation: DPO

More on this in your first assignment!!

Your first assignment might also need-

<https://huggingface.co/learn/deep-rl-course/en/unit0/introduction>
<https://github.com/vwxyzjn/cleanrl>

Chapter 4

Don't panic!! You will be taught these stuffs very soon.

What is Model Merging?

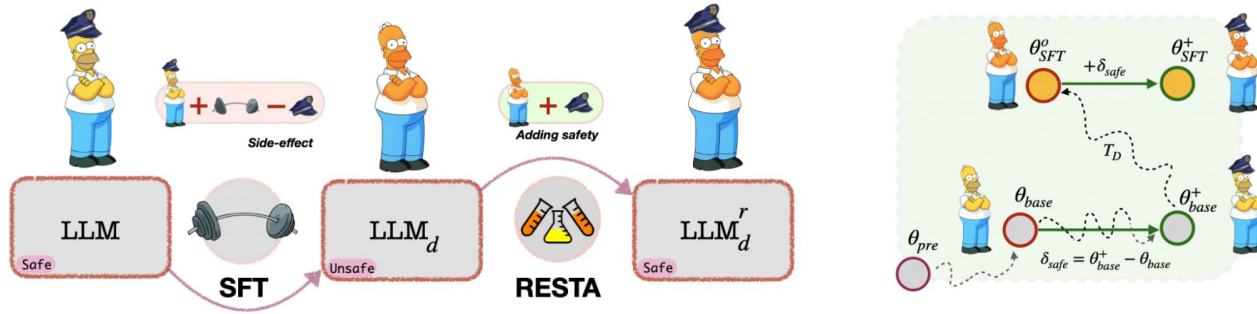


Figure: (Picked from RESTA paper) Representation of a brief idea of RESTA based safety decoding in LLMs

Why is Model Merging Required?

It's an exercise for you :)
Look out for some more tricky use cases.

Model Arithmetic

A quick go through the code -

<https://github.com/arcee-ai/mergekit>

Model Merging Methods

<https://github.com/arcee-ai/mergekit?tab=readme-ov-file#merge-methods>

Example – Dissecting Deeper Into Code

More on this in your second assignment!!

In-Context Learning

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

<https://medium.com/data-science/in-context-learning-approaches-in-large-language-models-9c0c53b116a1>

Function Vectors

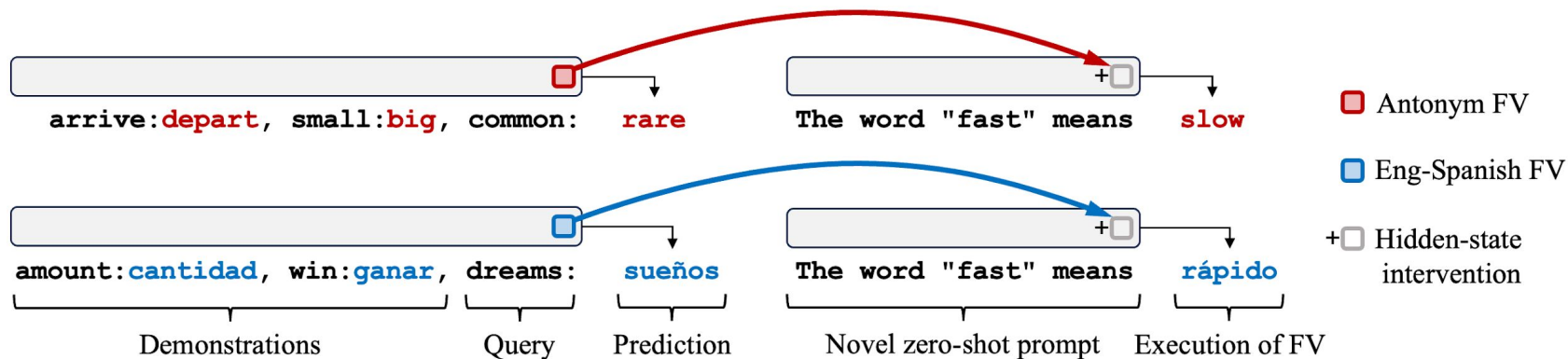


Figure:

https://github.com/ericwtodd/function_vectors?tab=readme-ov-file

Example – PyTorch Hooks [Needed for Function Vectors]

More on this in your second assignment!!

<https://www.digitalocean.com/community/tutorials/pytorch-101-understanding-graphs-and-automatic-differentiation>

<https://www.digitalocean.com/community/tutorials/pytorch-hooks-gradient-clipping-debugging>

THANK YOU